TCP congestion control in action and Introduction to socket programming

Part - 1

Let's now examine the amount of data sent per unit time from the client to the server. Rather than (tediously!) Calculating this from the raw data in the Wireshark window, we'll use one of Wireshark's TCP graphing utilities–Time-Sequence-Graph(Stevens)–to plot out data.

- Select a client-sent TCP segment in the Wireshark's "listing of captured-packets" window corresponding to the transfer of alice.txt from the client to gaia.cs.umass.edu.
- Then select the menu: Statistics->TCP Stream Graph-> Time-Sequence-Graph(Stevens2).
- You should see a plot that looks similar to the plot in Figure 5, which was created from the captured packets in the packet trace tcp-wireshark-trace1-1.
- You may have to expand, shrink, and fiddle around with the intervals shown in the axes in order to get your graph to look like Figure 5.

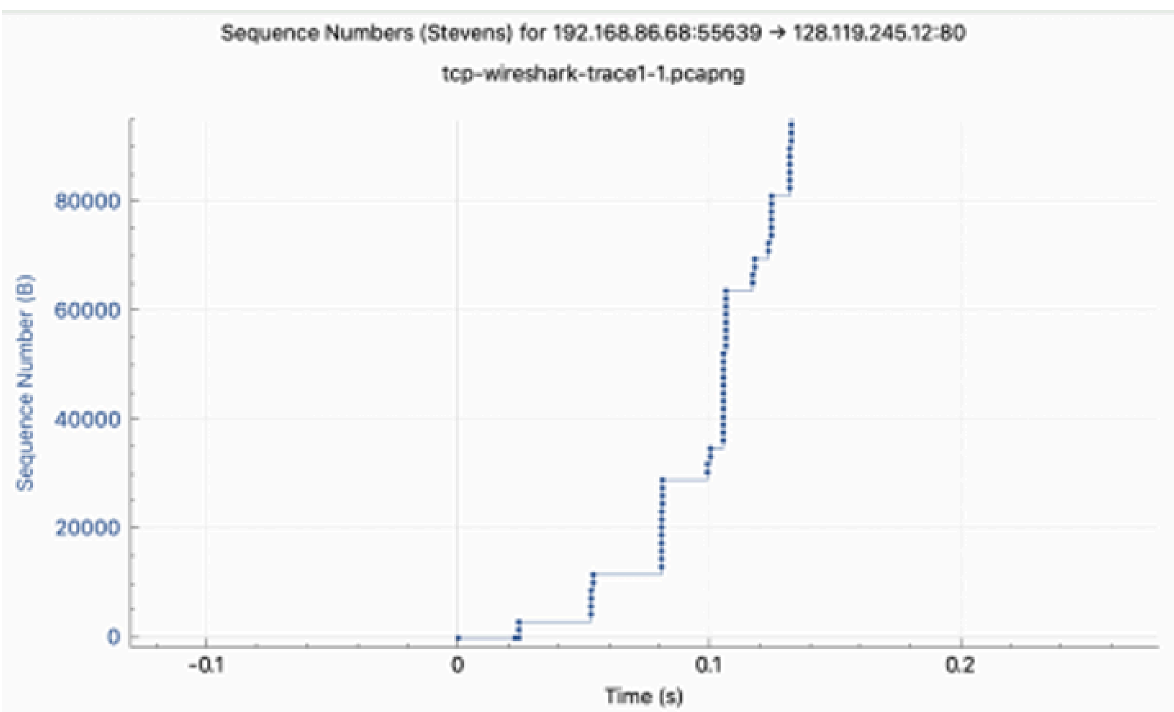

Figure 5: A sequence-number-versus-time plot (Stevens format) of TCP segments.

Here, each dot represents a TCP segment sent, plotting the sequence number of the segment versus the time at which it was sent. Note that a set of dots stacked above each other represents a series of packets (sometimes called a “fleet” of packets) that were sent back-to-back by the sender.

1. Use the Time-Sequence-Graph(Stevens) plotting tool to view the sequence number versus time plot of segments being sent from the client to the gaia.cs.umass.edu server. Consider the “fleets” of packets sent around t = 0.025, t = 0.053, t = 0.082 and t = 0.1. Comment on whether this looks as if TCP is in its slow start phase, congestion avoidance phase or some other phase. Figure 6 shows a slightly different view of this data.

2. These “fleets” of segments appear to have some periodicity. What can you say about the period?

3. Answer each of two questions above for the trace that you have gathered when you transferred a file from your computer to gaia.cs.umass.edu

Part - 2

Write a basic program for socket programming between a client and a server in preferred language.